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# HACKATHON 2026

*Idea Submission Template / Code. Create. Innovate.*

Team Name

Team Raksha

Problem Statement No.

2

Problem Statement Title

Improving Safety at Large Public Events

Team Leader Name

Shounak Joshi

# IDEA TITLE

## Know the crowd. Catch the bottleneck. Stop the surge.

Cameras measure how dense each zone is, and how fast it moves.  
It warns guards before a crush builds.

18

tracked at once

<1s

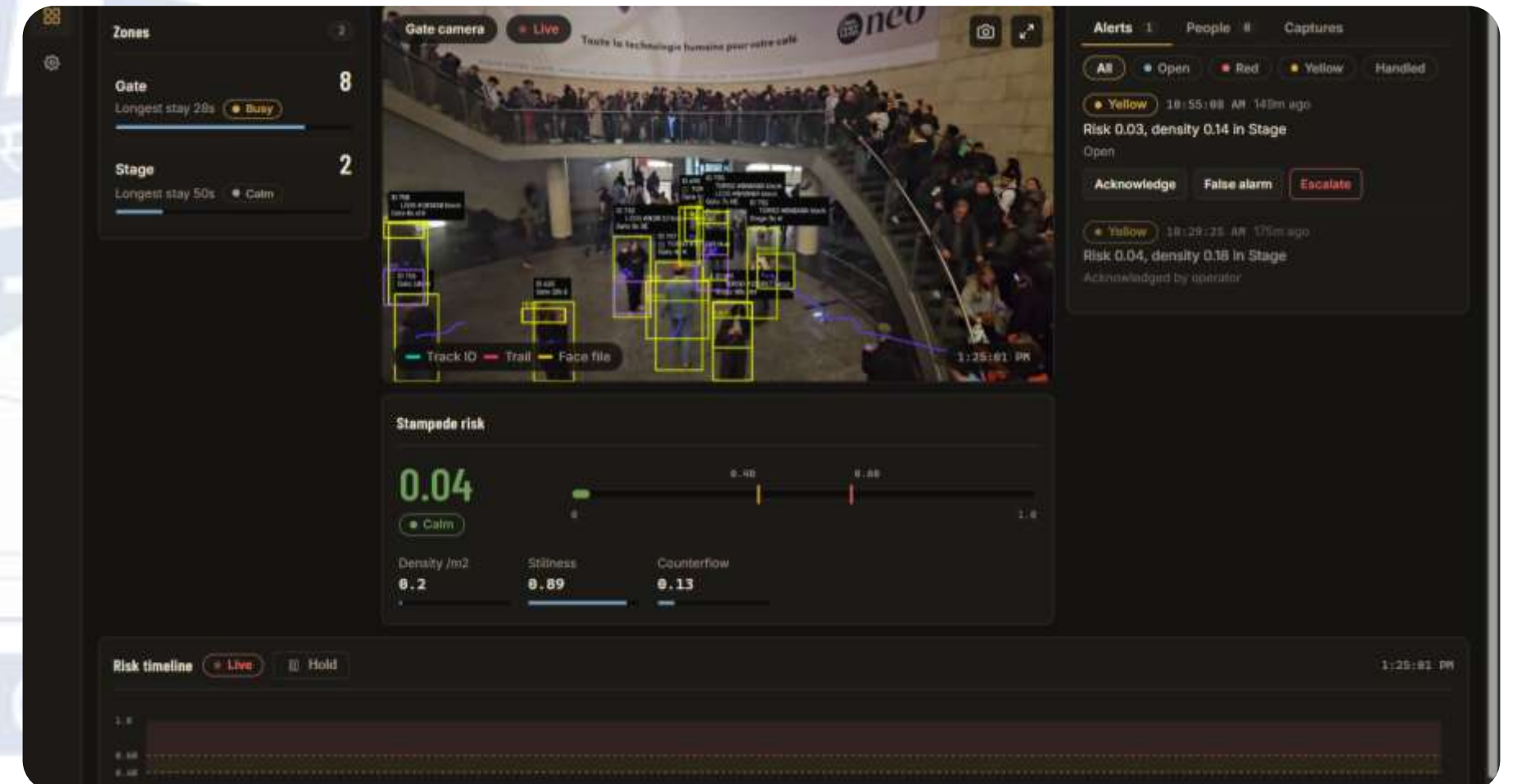
guard warned

15

video checks every  
second

### Watch it work

- 1 Live feed: every person counted, zone density tracked.
- 2 A zone packs up and stops: bottleneck flagged.
- 3 Guard warned in seconds; wall shows time left to critical.



The live wall, running on real station footage.

### What the colours mean



**Calm**

All is well.



**Watch**

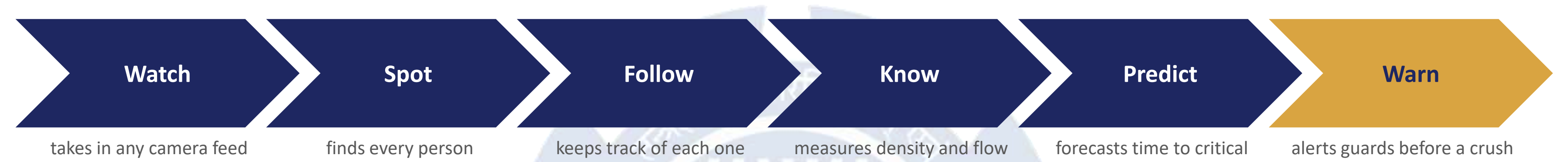
Getting crowded. Guards get an early warning.



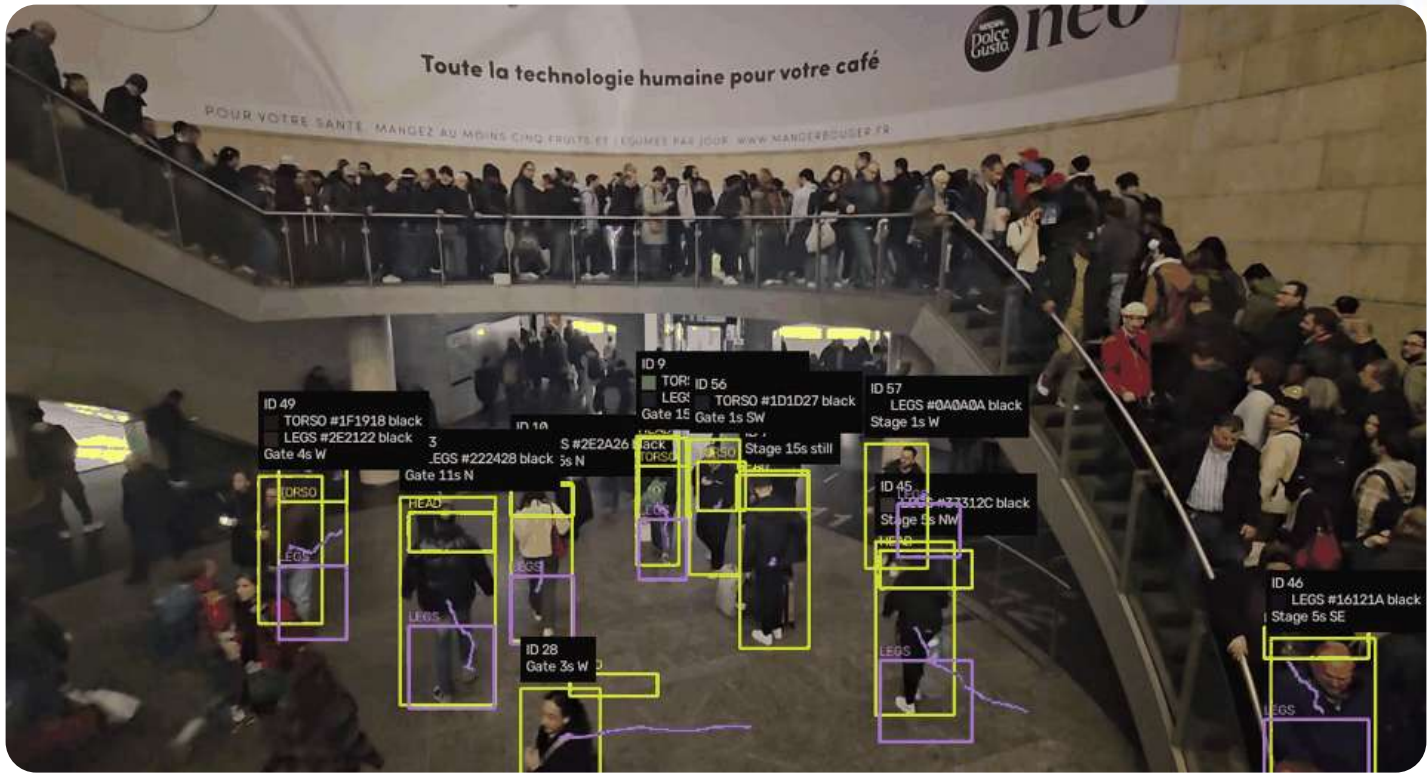
**Critical**

Crowd packed and stuck. Clear the area now.

# TECHNICAL APPROACH



In plain words: cameras share one venue map, the software flags where the crowd stops moving, and it warns guards before a crush builds.



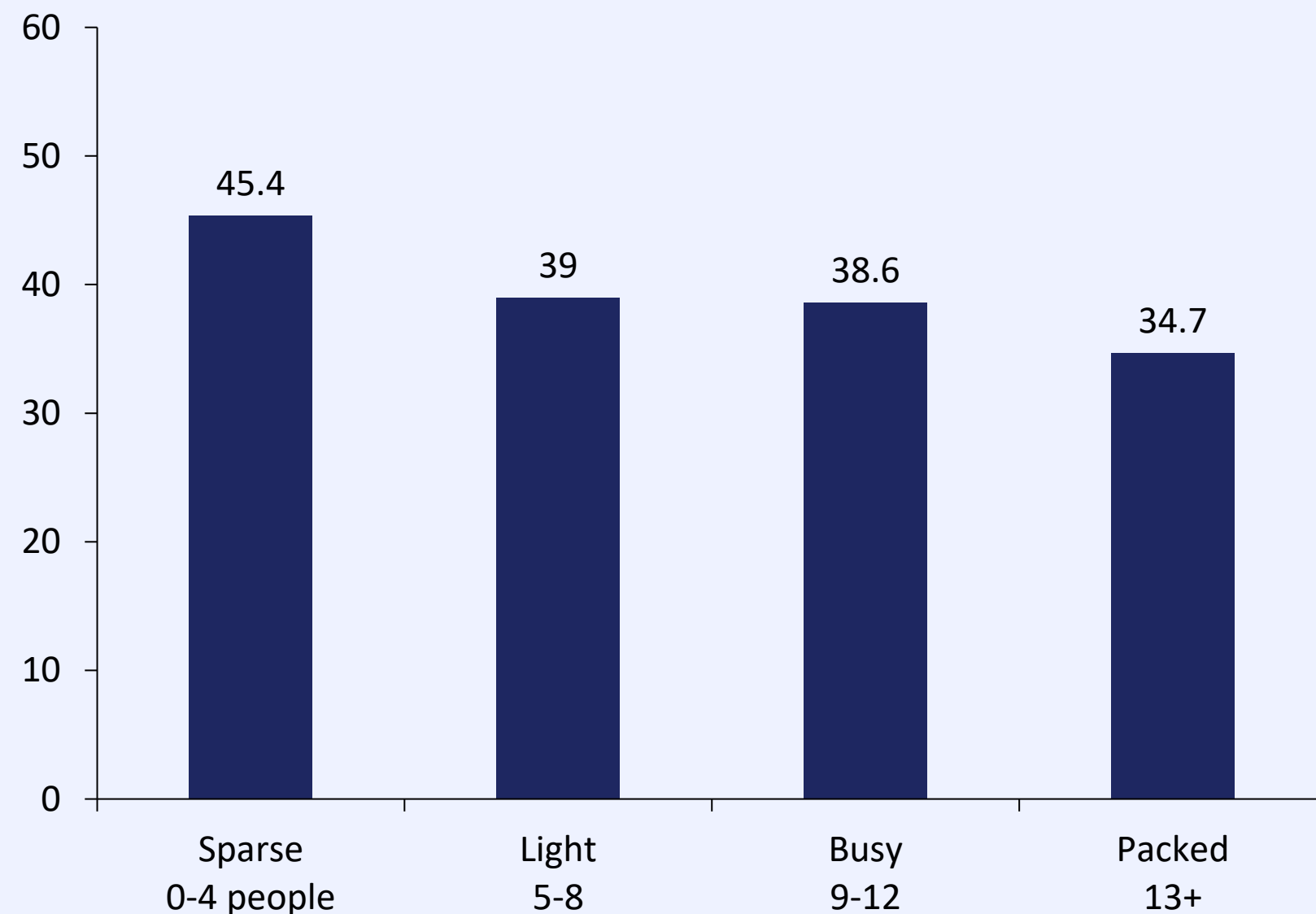
**Crowd metrics, not faces**  
Risk = density x stillness x counterflow. Stillness only escalates risk when the zone is also packed.

**Tested on real video**  
Followed 18 people at once across 2 minutes of busy station video.

**Works with normal cameras**  
Cameras link into one venue graph, so a jam is traced back to where it started. High, wide angles measure density best.

# FEASIBILITY AND VIABILITY

Checks per second as the crowd grows



Model and tracker time only, same laptop. FPS stays above 30 even when the frame fills with people.

## Forecast

The trend is fitted live, so the wall shows seconds to critical before the jam locks.

## Red alert

Fires when a zone packs tight and stops. The wall counts down the seconds to critical.

## Built to scale out

One laptop runs a venue today; an edge box per camera scales to mass gatherings.



# IMPACT AND BENEFITS

## When the alarm goes up

### Yellow watch

Guards keep eyes on the zone. Crowd flow is slowed.

### Red alert

Guards get steps on their handsets; a human confirms before any PA or door release.

### Handover

A sealed evidence pack goes to the police.

**Pilot targets: tens of seconds forecast before critical, guard warned under 1 second, every step logged.**

### Before

A wall of screens, one tired guard, and risks get missed.

### After

**One screen ranks zones and traces jams to their origin: security moves from reactive to proactive.**

### Organizers

Spot a growing jam early, and tell guards which zone to clear.

### Police

Every alert, zone and action exports as one evidence pack in a click.

### Public

Faces stay blurred in saved video.

# RESEARCH AND REFERENCES

## The science behind it

### Finding and following people

Finds every person and follows each one. Uses YOLOv8 and ByteTrack, the standard real-time pairing for crowd tracking.

Code: [github.com/ultralytics/ultralytics](https://github.com/ultralytics/ultralytics), [github.com/ifzhang/ByteTrack](https://github.com/ifzhang/ByteTrack)

### Phase 2: face search

Not in the MVP. The pipeline is modular, so agencies can plug in their own face-search API later for post-incident cases (opt-in, B2G).

Code: [github.com/deepinsight/insightface](https://github.com/deepinsight/insightface), [github.com/facebookresearch/faiss](https://github.com/facebookresearch/faiss)

### Predicting crushes

The AURORA warning model plus our live optical-flow stall meter spot danger early. A Delhi trial cut response time by 37 percent.

Ideas: AURORA warning model, optical-flow stall metrics, STAR-Crowd Delhi trial

### Why India needs it

Packed crowds and blocked exits turned deadly at Hathras, Kumbh and railway stations.

Test video: [Wikimedia Commons](#) public footage (station, crossing, temple)

## Privacy built in

### Anonymized by default

Saved video is face-blurred. Live alerts show zones and counts, not names.

### 7-day routine wipe

Daily footage wipes after 7 days. Flagged incidents are exported first and kept for the police.

### Tamper-evident log

Every action is hash-chained. Any edit leaves a trace.

### No ID in core

Crowd mode runs with zero identity. Face search stays opt-in Phase 2 only.